

Software Requirements Specification (SRS)

Invoice Processing — From PDF to Decision | Version 1.0

1. System Scope

The system converts a vendor invoice PDF into a controlled business decision. The system is not the system of record for procurement or payment; the MVP uses seeded vendor and PO data and treats them as authoritative reference inputs.

2. Roles & Permissions

Role	Permissions
AP Analyst	Upload/process invoices; view runs; inspect evidence; resolve REVIEW; select PO candidate; add review notes
AP Manager	All analyst permissions; override final decision; configure demo tolerance policy
Admin	Load/replace vendor and PO fixture data; configure model provider; inspect system health
Auditor	Read-only access to runs, evidence, rule results, decisions, and override history

3. Functional Requirements

ID	Requirement	Testable Requirement
FR-01	Invoice upload	System shall accept PDF invoices up to configurable size and reject unsupported files before processing.
FR-02	Run creation	System shall create a unique run ID and timestamp before model processing.
FR-03	PDF preparation	System shall render pages/images for multimodal analysis without treating OCR text extraction as the primary document-understanding method.
FR-04	Structured extraction	System shall extract vendor, invoice number, invoice date, due date, currency, subtotal, tax, total, PO/reference numbers, payment details, and line items where present.
FR-05	Evidence	Each extracted field shall retain a source/evidence representation where technically available and a field-level confidence/status.
FR-06	Extraction verification	System shall run consistency checks for required fields and arithmetic and may call a second model verification step for uncertain fields.
FR-07	Vendor resolution	System shall resolve a vendor using deterministic identifiers first and fuzzy/semantic evidence second.
FR-08	Candidate generation	System shall return zero, one, or multiple PO candidates and shall not require an explicit PO number to continue.
FR-09	Candidate scoring	System shall score candidates using

		multiple signals and record each signal contribution.
FR-10	PO validation	System shall validate vendor, PO status, currency, quantities, prices, subtotal/tax/total, tolerance, and remaining PO balance.
FR-11	Split invoice support	System shall support multiple invoices consuming the same PO by maintaining remaining PO quantity/value in the fixture state.
FR-12	Duplicate detection	System shall check exact invoice number/vendor duplicates and content/amount/date similarity.
FR-13	Decision engine	System shall return APPROVE, REVIEW, or REJECT using deterministic policy rules.
FR-14	Explainability	System shall return reason codes and human-readable reasons tied to rule/evidence results.
FR-15	Run timeline	System shall expose stage-level progress and status.
FR-16	History	System shall persist completed and failed runs and allow filtering by decision/status/vendor/date.
FR-17	Manual review	System shall allow an authorized reviewer to choose a PO or override a decision with a mandatory reason.
FR-18	Audit trail	System shall record actor, timestamp, old decision, new decision, reason, and relevant evidence for manual overrides.

4. Business Rules

ID	Rule	Requirement
BR-01	Critical missing data	Missing invoice total, currency, or vendor identity shall prevent APPROVE and route to REVIEW unless policy explicitly supports a safe derivation.
BR-02	Vendor mismatch	Invoice vendor and PO vendor must match an approved relationship; a PO number alone cannot override a vendor mismatch.
BR-03	PO status	Closed/cancelled PO shall not be auto-approved.
BR-04	Currency	Invoice and PO currency must match unless an explicitly configured conversion rule exists.
BR-05	Amount tolerance	Invoice amount may differ from PO/remaining amount only within configured absolute/percentage tolerance.
BR-06	Quantity	Invoiced quantity shall not exceed remaining PO quantity for a matched line unless an authorized exception is configured.
BR-07	Split invoices	A PO may be consumed by multiple valid invoices; validation compares against remaining quantity/value, not only original PO amount.
BR-08	Duplicate	Exact duplicate invoice number for the same vendor or high-confidence content

		duplicate shall prevent APPROVE.
BR-09	Candidate ambiguity	If top PO score is below minimum or the margin to the runner-up is below threshold, return REVIEW.
BR-10	Model uncertainty	Low extraction confidence on a financially material field shall route to REVIEW.
BR-11	Arithmetic	Subtotal + tax and line extensions shall reconcile within configured rounding tolerance.
BR-12	Negative invoices	Credit notes/negative invoices shall route to a separate review policy in MVP.
BR-13	Final authority	The model shall never be the final authority for approval/rejection; deterministic policy evaluates validated facts.

5. PO Matching Algorithm

Candidate generation:

1. If explicit PO number exists, retrieve exact PO(s) and verify vendor/status/currency; also retain a small alternative candidate set if the reference is ambiguous.
2. If no PO number exists, filter by vendor and currency, then rank by amount/remaining balance, date proximity, item identifiers and semantic line similarity.
3. Use fuzzy matching for normalized vendor/reference text and semantic similarity for descriptions only as supporting evidence.
4. Return top K candidates with a transparent score breakdown.

Illustrative scoring model (weights are configurable for the demo):

Signal	Weight	Example scoring
Exact PO number	0.30	1.0 exact, 0 otherwise
Vendor match	0.20	1.0 approved exact/normalized; lower for fuzzy
Amount compatibility	0.15	Higher when within tolerance/remaining balance
Line/SKU overlap	0.15	Normalized item-code/description overlap
Quantity compatibility	0.08	1.0 if within remaining quantity
Reference/date evidence	0.05	Supporting match
Semantic similarity	0.07	Embedding/model similarity; never sufficient alone

Decision thresholds are policy parameters, not model outputs. Example: auto-match only when score ≥ 0.85 and the top-vs-runner-up margin ≥ 0.10 , with all hard controls passing. Cases below that threshold become REVIEW.

6. Data Requirements

Entity	Required Fields
Invoice	run_id, file_hash, vendor_name, invoice_number, invoice_date, due_date, currency, subtotal, tax, total, po_reference, lines[], extraction_status
InvoiceLine	description, item_code, quantity, unit_price, tax_rate, line_total
Vendor	vendor_id, legal_name, aliases, tax_id, currency, status
PO	po_id, vendor_id, currency, status, issue_date, total_value, remaining_value, lines[]

POLine	line_id, item_code, description, ordered_qty, invoiced_qty, remaining_qty, unit_price, line_total
Run	run_id, timestamps, stage statuses, decision, reason_codes, model metadata
DecisionAudit	run_id, actor, action, old_value, new_value, reason, timestamp

7. Validation Requirements

- File validation: MIME/type, extension, size, page count, corrupted PDF handling.
- Schema validation: model response must conform to JSON schema; malformed responses are rejected and retried once.
- Arithmetic validation: line totals, subtotal, tax and grand total are independently recalculated.
- Reference validation: vendor and PO IDs must exist in reference data.
- Tolerance validation: compare invoice value to remaining PO value, not only original PO value.
- Duplicate validation: normalized invoice number + vendor, file hash, and high-similarity invoice attributes.
- Confidence validation: material fields below threshold cannot silently pass.
- Decision validation: APPROVE is impossible when any critical blocking rule is failed.

8. Authentication & Authorization

- MVP may use simple authenticated demo accounts with role claims.
- Passwords must never be stored in plaintext; use a standard password hash or external auth provider.
- Backend shall enforce authorization; hiding a UI control is not a security boundary.
- API/model keys shall remain server-side and never be embedded in frontend code.
- Reviewer overrides require an authenticated role and mandatory reason.

9. Error Handling

Failure	System Behavior
Invalid PDF	Reject before processing; show actionable message
Model timeout	Retry once; then mark stage failed and stop financial decision
Invalid model JSON	Retry with schema-constrained prompt; then REVIEW/ERROR
No PO candidate	Continue to decision as REVIEW with reason NO_PO_MATCH
Multiple PO candidates	Continue; show candidates; REVIEW if ambiguity remains
Reference data unavailable	Do not approve; show dependency error
Duplicate detected	Do not approve; show duplicate evidence
Arithmetic mismatch	Do not approve; show exact discrepancy
Unexpected exception	Persist failed run and user-safe error; no silent approval

10. Security Requirements

- Use HTTPS in deployed environments.
- Validate and sandbox file handling; never execute uploaded content.
- Store only required invoice data; use access controls around financial documents.
- Redact secrets from logs.
- Record model/provider and prompt-version metadata for reproducibility without exposing secrets.
- Use immutable or append-only audit records for decisions and overrides in a production implementation.
- Treat external model providers as data processors; for real deployment, verify retention/privacy terms before sending financial documents.

11. Performance Requirements

- UI shall show stage progress immediately after upload.
- Target end-to-end MVP run: ≤ 30 seconds excluding unavoidable provider/network variance.
- Candidate retrieval from a small seeded PO dataset should be near-instantaneous.
- The backend should avoid sending the full PO database to the model.
- Large PDFs should be page-limited or configurable to avoid unnecessary model cost.

12. Acceptance Test Matrix

Test	Input	Expected
AT-01 Happy path	Clean invoice + exact PO	APPROVE
AT-02 Missing PO	Vendor + amount uniquely identifies PO	APPROVE if all hard controls pass
AT-03 Wrong vendor	PO number belongs to another vendor	REVIEW/REJECT; never APPROVE
AT-04 Split invoice	Second invoice consumes remaining PO balance	APPROVE if remaining balance passes
AT-05 Over tolerance	Invoice exceeds configured tolerance	REVIEW or REJECT
AT-06 Duplicate	Same vendor + invoice number already processed	REJECT/duplicate block
AT-07 Near duplicate	Different filename, same invoice content	Duplicate warning; no APPROVE
AT-08 Ambiguous PO	Two plausible candidates	REVIEW with both candidates
AT-09 Closed PO	Invoice references closed PO	REJECT/REVIEW per policy
AT-10 Bad scan	Low-confidence critical fields	REVIEW
AT-11 Currency mismatch	USD invoice, EUR PO	REVIEW/REJECT
AT-12 Arithmetic mismatch	Line/tax/total inconsistency	REVIEW/REJECT